

Name :

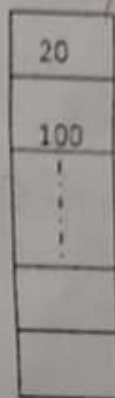
ID :

Wed 10-12 model (H)

stack.h

```
1 #define Max 20
2 typedef int StackType [ Max];
3 //precondition : none.
4 //postcondition : The stack is initialized to be empty.
5 void create (StackType *S);
6 // precondition : The stack is not empty.
7 //postcondition: An item removed from the stack.
8 void pop (int *, StackType *);
9 int StackEmpty (StackType );
10 int StackFull (StackType );
```

Max-1



20 pushed first then 100

The index of the top element is kept in the first cell of the array, and the function create is written as follow:

```
void create(StackType *S){ (*S)[0] = Max-1 }
```

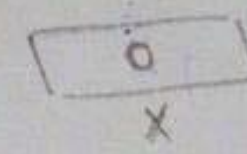
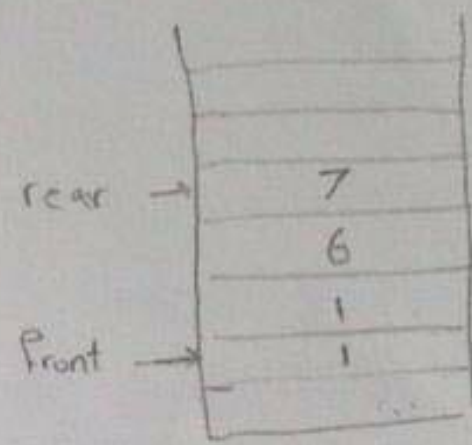
Question: Write a pop function with the same header as in stack.h.

```

X = 0;
Y = 1;
Enqueue (X, &Queue);
Enqueue (Y, &Queue);
Enqueue (Y, &Queue);
Z = Y + 5; 6
Enqueue (Z, &Queue);
Enqueue (7, &Queue);
Dequeue (&X, &Queue);
Enqueue (Y, &Queue);
while (! QueueEmpty (Queue)) {
    Dequeue (&Z, &Queue);
    printf("%d \n", Z); }

```

1
1
6
7
1



B) Given the following specification of the Pop operation:

*void Pop(Stack *S, StackEntry *E, int * UnderFlow)*

Function: Remove top element from a stack and return it.

Precondition: A stack has been created

Postcondition: If stack is empty UnderFlow variable will have the value 1
 Otherwise UnderFlow variable will have the value 0, the top element of the stack will be deleted, and its value will be returned using E variable

Write a function to implement this operation using the linked stack implementation.

*void Pop (Stack *Ps) {*

ATTEMPT IN ALL QUESTIONS:

Question 1:

A) Trace the following segment of code and show the output:

```
CreateQueue(&Queue);  
X = 0;  
Y = 1;  
Enqueue (X, &Queue);  
Enqueue (Y, &Queue);  
Enqueue (Y, &Queue);  
Z = Y + 5; 6  
Enqueue (Z, &Queue);  
Enqueue (7, &Queue);  
Dequeue (&X, &Queue);  
Enqueue (Y, &Queue);  
while (! QueueEmpty (Queue)) {  
    Dequeue (&Z, &Queue);  
    printf("%d \n", Z); }
```

